

Optimizing-Email-Campaign-Performance-to-Boost-Engagement-Account-Linking-and-Funding-Rates

Executive Summary

This project analyzes a large-scale email campaign experiment (480,000 users) to optimize **traffic, engagement, conversion, and customer retention performance**.

By integrating campaign logs, user behavior data, and control benchmarks, I evaluated:

- **Traffic & engagement** (open rate, send-order performance)
- **Conversion performance** (account linking rate, funding rate)
- **Retention & activity signals**
- **Campaign ROI impact through A/B testing**
- **Funnel efficiency from engagement → linking → funding**

The analysis delivered statistically validated improvements in conversion performance and actionable recommendations for targeting, cadence optimization, and campaign strategy refinement.

Business Context (E-commerce Framing)

This project simulates a real-world e-commerce growth analytics environment where:

- Email campaigns drive **traffic**
- User engagement drives **conversion**
- Conversion leads to **revenue events (funding)**

- Sustained activity signals reflect **retention**
 - Campaign strategy must balance performance lift with churn risk
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KPI Framework

E-commerce KPI	Project Metric
Traffic	Email delivery volume
Engagement	Open rate
Conversion	Link rate, Funding rate
Revenue Proxy	Funding conversion uplift
Retention	Post-campaign activity rate
Campaign ROI	Treatment vs Control lift
Customer Risk	Unsubscribe & Spam rate

What I Did

1. Analyze e-commerce performance metrics (traffic, conversion, revenue, retention)

- Calculated open rate across 10 campaign messages
- Measured link rate and funding rate across 24 segment-frequency combinations
- Built conversion funnel:

Open Rate → Link Rate → Funding Rate

- Quantified incremental lift vs control groups using statistical testing

- Identified high-value customer segments based on behavioral signals
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2. Develop dashboards and reports to track KPIs

Built analytical views including:

- Open rate ranking dashboard (content performance)
- Segment-level engagement heatmaps
- Treatment vs control lift comparison bar charts
- Time-series engagement dashboard (send order 0–9)
- Funnel performance visualization
- Negative event monitoring table (unsubscribe/spam)

Each dashboard answers a specific business question and supports campaign decision-making.

3. Monitor customer behavior, funnel performance, and engagement

- Reconstructed per-user time-series engagement using randomized send order
- Identified engagement decay pattern:
 - Peak open rate at Day 1–2
 - Decline in later sends
- Diagnosed campaign fatigue effects
- Quantified drop-off between:
 - Engagement and linking

- Linking and funding

4. Conduct product/promotion analysis (campaign content testing)

- Compared 10 different email content variants
- Identified highest-performing content ([ml_funding_faq](#), ~26% open rate)
- Ranked content by performance
- Evaluated negative feedback impact per content variant

This mimics promotional performance optimization in merchandising.

5. Partner with marketing to analyze campaign performance & ROI

- Measured incremental funding lift vs control
- Conducted one-sided proportion z-tests
- Calculated:
 - Δ funding rate
 - Δ link rate
 - Statistical significance

Most segments showed statistically significant funding uplift.

This directly supports campaign ROI justification.

6. Support A/B testing and experimentation

- Compared 12 treatment segments against baseline controls
 - Validated statistical assumptions
 - Implemented hypothesis testing framework:
H0: Treatment $\leq$ Control
H1: Treatment $>$ Control
 - Flagged statistically significant results
 - Identified segments where lift was not significant
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7. Ensure data accuracy & consistency

- Detected delivery anomaly (near-zero delivery on order_9 weekly cohort)
- Identified increasing NaN rates over campaign duration
- Flagged experiment instrumentation issues that could bias results
- Recommended excluding corrupted data before final rollout decisions

This demonstrates governance and analytical integrity.

8. Provide insights and recommendations to stakeholders

Key Recommendations

- Prioritize high-performing content (funding FAQ)
- Target recently active users for highest ROI
- Shorten campaign duration (engagement decays after Day 2–3)
- Limit frequency to reduce unsubscribe/spam risk

- Fix delivery tracking issues before scaling
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Technical Implementation

- Python (Pandas, NumPy)
 - Statsmodels (proportion z-test)
 - Matplotlib & Seaborn
 - Large-scale data manipulation (480K users)
 - Segmentation analysis
 - Funnel modeling
 - Statistical validation
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Advanced Analytical Components

- ✓ Correlation heatmaps (engagement drivers)
 - ✓ Heterogeneous treatment effect analysis
 - ✓ Funnel delta modeling
 - ✓ Extra conversion rate metric
 - ✓ Negative event monitoring
 - ✓ Time-series send-order reconstruction
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Impact Summary

- Identified top-performing content with 26% open rate

- Demonstrated statistically significant funding uplift across most segments
 - Diagnosed engagement decay pattern to optimize cadence
 - Balanced performance lift with long-term customer retention risk
 - Built reusable KPI tracking framework for campaign analytics
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Why This Project Is Relevant for E-commerce Analytics Roles

This project demonstrates:

- Performance metric analysis
 - Funnel optimization
 - Campaign ROI evaluation
 - Customer segmentation strategy
 - A/B testing execution
 - Data quality governance
 - Stakeholder-ready visualization
 - Actionable business recommendations
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